

Model for “Mn3Sn”

Generated on 2026-08-07 20:26:49 by MultiPie 2.3.2

General Condition

- Basis type: 1g
- SAMB selection:
 - Type: [Q, G]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_{1g} , E_{1u} , E_{2g} , E_{2u}]
 - Spin (s): [0, 1]
- Atomic selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_{1g} , E_{1u} , E_{2g} , E_{2u}]
 - Spin (s): [0, 1]
- Site-cluster selection:
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_{1g} , E_{1u} , E_{2g} , E_{2u}]
- Bond-cluster selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_{1g} , E_{1u} , E_{2g} , E_{2u}]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: **false**

Group and Unit Cell

- Group: SG No. 194 D_{6h}^4 $P6_3/mmc$ [hexagonal]
- Associated point group: PG No. 194 D_{6h} $6/mmm$ [hexagonal]
- Unit cell:
 - $a = 1.00000$, $b = 1.00000$, $c = 1.00000$, $\alpha = 90.0$, $\beta = 90.0$, $\gamma = 120.0$
- Lattice vectors (conventional cell):
 - $\mathbf{a}_1 = [1.00000, 0.00000, 0.00000]$
 - $\mathbf{a}_2 = [-0.50000, 0.86603, 0.00000]$
 - $\mathbf{a}_3 = [0.00000, 0.00000, 1.00000]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1 0\}$	2	$\{3_{001}^+ 0\}$	3	$\{3_{001}^- 0\}$	4	$\{2_{001} 00\frac{1}{2}\}$	5	$\{6_{001}^- 00\frac{1}{2}\}$
6	$\{6_{001}^+ 00\frac{1}{2}\}$	7	$\{2_{110} 0\}$	8	$\{2_{100} 0\}$	9	$\{2_{010} 0\}$	10	$\{2_{1-10} 00\frac{1}{2}\}$
11	$\{2_{120} 00\frac{1}{2}\}$	12	$\{2_{210} 00\frac{1}{2}\}$	13	$\{-1 0\}$	14	$\{-3_{001}^+ 0\}$	15	$\{-3_{001}^- 0\}$
16	$\{m_{001} 00\frac{1}{2}\}$	17	$\{-6_{001}^- 00\frac{1}{2}\}$	18	$\{-6_{001}^+ 00\frac{1}{2}\}$	19	$\{m_{110} 0\}$	20	$\{m_{100} 0\}$
21	$\{m_{010} 0\}$	22	$\{m_{1-10} 00\frac{1}{2}\}$	23	$\{m_{120} 00\frac{1}{2}\}$	24	$\{m_{210} 00\frac{1}{2}\}$		

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_{1g})$	A_{1g}	0	Q, T	-	-	1
2	$\mathbb{Q}_2(A_{1g})$	A_{1g}	2	Q, T	-	-	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
3	$\mathbb{Q}_4(A_{1g})$	A_{1g}	4	Q, T	-	-	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
4	$\mathbb{Q}_6(A_{1g}, 2)$	A_{1g}	6	Q, T	2	-	$\frac{\sqrt{462}(x-y)(x+y)(x^2-4xy+y^2)(x^2+4xy+y^2)}{32}$
5	$\mathbb{G}_0(A_{1u})$	A_{1u}	0	G, M	-	-	1
6	$\mathbb{G}_2(A_{1u})$	A_{1u}	2	G, M	-	-	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
7	$\mathbb{G}_4(A_{1u})$	A_{1u}	4	G, M	-	-	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
8	$\mathbb{G}_6(A_{1u}, 2)$	A_{1u}	6	G, M	2	-	$\frac{\sqrt{462}(x-y)(x+y)(x^2-4xy+y^2)(x^2+4xy+y^2)}{32}$
9	$\mathbb{Q}_7(A_{1u})$	A_{1u}	7	Q, T	-	-	$\frac{\sqrt{6006}xyz(x^2-3y^2)(3x^2-y^2)}{16}$
10	$\mathbb{G}_1(A_{2g})$	A_{2g}	1	G, M	-	-	z
11	$\mathbb{G}_3(A_{2g})$	A_{2g}	3	G, M	-	-	$-\frac{z(3x^2+3y^2-2z^2)}{2}$
12	$\mathbb{G}_5(A_{2g})$	A_{2g}	5	G, M	-	-	$\frac{z(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4)}{8}$
13	$\mathbb{Q}_6(A_{2g})$	A_{2g}	6	Q, T	-	-	$\frac{\sqrt{462}xy(x^2-3y^2)(3x^2-y^2)}{16}$
14	$\mathbb{Q}_1(A_{2u})$	A_{2u}	1	Q, T	-	-	z
15	$\mathbb{Q}_3(A_{2u})$	A_{2u}	3	Q, T	-	-	$-\frac{z(3x^2+3y^2-2z^2)}{2}$
16	$\mathbb{G}_6(A_{2u})$	A_{2u}	6	G, M	-	-	$\frac{\sqrt{462}xy(x^2-3y^2)(3x^2-y^2)}{16}$
17	$\mathbb{Q}_7(A_{2u}, 2)$	A_{2u}	7	Q, T	2	-	$\frac{\sqrt{6006}z(x-y)(x+y)(x^2-4xy+y^2)(x^2+4xy+y^2)}{32}$
18	$\mathbb{G}_3(B_{1g})$	B_{1g}	3	G, M	-	-	$\frac{\sqrt{10}y(3x^2-y^2)}{4}$
19	$\mathbb{Q}_4(B_{1g})$	B_{1g}	4	Q, T	-	-	$\frac{\sqrt{70}xz(x^2-3y^2)}{4}$
20	$\mathbb{Q}_6(B_{1g})$	B_{1g}	6	Q, T	-	-	$-\frac{\sqrt{210}xz(x^2-3y^2)(3x^2+3y^2-8z^2)}{16}$
21	$\mathbb{Q}_3(B_{1u})$	B_{1u}	3	Q, T	-	-	$\frac{\sqrt{10}y(3x^2-y^2)}{4}$
22	$\mathbb{G}_4(B_{1u})$	B_{1u}	4	G, M	-	-	$\frac{\sqrt{70}xz(x^2-3y^2)}{4}$
23	$\mathbb{Q}_5(B_{1u})$	B_{1u}	5	Q, T	-	-	$-\frac{\sqrt{70}y(3x^2-y^2)(x^2+y^2-8z^2)}{16}$
24	$\mathbb{Q}_9(B_{1u}, 1)$	B_{1u}	9	Q, T	1	-	$\frac{\sqrt{24310}y(3x^2-y^2)(3x^6-27x^4y^2+33x^2y^4-y^6)}{256}$
25	$\mathbb{G}_3(B_{2g})$	B_{2g}	3	G, M	-	-	$\frac{\sqrt{10}x(x^2-3y^2)}{4}$
26	$\mathbb{Q}_4(B_{2g})$	B_{2g}	4	Q, T	-	-	$\frac{\sqrt{70}yz(3x^2-y^2)}{4}$
27	$\mathbb{Q}_6(B_{2g})$	B_{2g}	6	Q, T	-	-	$-\frac{\sqrt{210}yz(3x^2-y^2)(3x^2+3y^2-8z^2)}{16}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
28	$\mathbb{Q}_3(B_{2u})$	B_{2u}	3	Q, T	-	-	$\frac{\sqrt{10}x(x^2-3y^2)}{4}$
29	$\mathbb{G}_4(B_{2u})$	B_{2u}	4	G, M	-	-	$\frac{\sqrt{70}yz(3x^2-y^2)}{4}$
30	$\mathbb{Q}_5(B_{2u})$	B_{2u}	5	Q, T	-	-	$-\frac{\sqrt{70}x(x^2-3y^2)(x^2+y^2-8z^2)}{16}$
31	$\mathbb{Q}_9(B_{2u}, 1)$	B_{2u}	9	Q, T	1	-	$\frac{\sqrt{24310}x(x^2-3y^2)(x^6-33x^4y^2+27x^2y^4-3y^6)}{256}$
32	$\mathbb{G}_{1,1}(E_{1g})$	E_{1g}	1	G, M	-	1	x
33	$\mathbb{G}_{1,2}(E_{1g})$					2	y
34	$\mathbb{Q}_{2,1}(E_{1g})$	E_{1g}	2	Q, T	-	1	$\sqrt{3}yz$
35	$\mathbb{Q}_{2,2}(E_{1g})$					2	$-\sqrt{3}xz$
36	$\mathbb{G}_{3,1}(E_{1g})$	E_{1g}	3	G, M	-	1	$\frac{\sqrt{6}x(-x^2-y^2+4z^2)}{4}$
37	$\mathbb{G}_{3,2}(E_{1g})$					2	$\frac{\sqrt{6}y(-x^2-y^2+4z^2)}{4}$
38	$\mathbb{Q}_{4,1}(E_{1g})$	E_{1g}	4	Q, T	-	1	$\frac{\sqrt{10}yz(-3x^2-3y^2+4z^2)}{4}$
39	$\mathbb{Q}_{4,2}(E_{1g})$					2	$\frac{\sqrt{10}xz(3x^2+3y^2-4z^2)}{4}$
40	$\mathbb{G}_{5,1}(E_{1g}, 1)$	E_{1g}	5	G, M	1	1	$\frac{3\sqrt{14}x(x^4-10x^2y^2+5y^4)}{16}$
41	$\mathbb{G}_{5,2}(E_{1g}, 1)$					2	$\frac{3\sqrt{14}y(-5x^4+10x^2y^2-y^4)}{16}$
42	$\mathbb{Q}_{6,1}(E_{1g}, 1)$	E_{1g}	6	Q, T	1	1	$\frac{3\sqrt{154}yz(5x^4-10x^2y^2+y^4)}{16}$
43	$\mathbb{Q}_{6,2}(E_{1g}, 1)$					2	$\frac{3\sqrt{154}xz(x^4-10x^2y^2+5y^4)}{16}$
44	$\mathbb{Q}_{6,1}(E_{1g}, 2)$	E_{1g}	6	Q, T	2	1	$\frac{\sqrt{21}yz(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4)}{8}$
45	$\mathbb{Q}_{6,2}(E_{1g}, 2)$					2	$\frac{\sqrt{21}xz(-5x^4-10x^2y^2+20x^2z^2-5y^4+20y^2z^2-8z^4)}{8}$
46	$\mathbb{Q}_{8,1}(E_{1g}, 1)$	E_{1g}	8	Q, T	1	1	$\frac{3\sqrt{715}yz(7x^6-35x^4y^2+21x^2y^4-y^6)}{32}$
47	$\mathbb{Q}_{8,2}(E_{1g}, 1)$					2	$\frac{3\sqrt{715}xz(-x^6+21x^4y^2-35x^2y^4+7y^6)}{32}$
48	$\mathbb{Q}_{1,1}(E_{1u})$	E_{1u}	1	Q, T	-	1	x

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
49	$\mathbb{Q}_{1,2}(E_{1u})$					2	y
50	$\mathbb{G}_{2,1}(E_{1u})$	E_{1u}	2	G, M	-	1	$\sqrt{3}yz$
51	$\mathbb{G}_{2,2}(E_{1u})$					2	$-\sqrt{3}xz$
52	$\mathbb{Q}_{3,1}(E_{1u})$	E_{1u}	3	Q, T	-	1	$\frac{\sqrt{6}x(-x^2-y^2+4z^2)}{4}$
53	$\mathbb{Q}_{3,2}(E_{1u})$					2	$\frac{\sqrt{6}y(-x^2-y^2+4z^2)}{4}$
54	$\mathbb{G}_{4,1}(E_{1u})$	E_{1u}	4	G, M	-	1	$\frac{\sqrt{10}yz(-3x^2-3y^2+4z^2)}{4}$
55	$\mathbb{G}_{4,2}(E_{1u})$					2	$\frac{\sqrt{10}xz(3x^2+3y^2-4z^2)}{4}$
56	$\mathbb{Q}_{5,1}(E_{1u}, 1)$	E_{1u}	5	Q, T	1	1	$\frac{3\sqrt{14}x(x^4-10x^2y^2+5y^4)}{16}$
57	$\mathbb{Q}_{5,2}(E_{1u}, 1)$					2	$\frac{3\sqrt{14}y(-5x^4+10x^2y^2-y^4)}{16}$
58	$\mathbb{Q}_{5,1}(E_{1u}, 2)$	E_{1u}	5	Q, T	2	1	$\frac{\sqrt{15}x(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4)}{8}$
59	$\mathbb{Q}_{5,2}(E_{1u}, 2)$					2	$\frac{\sqrt{15}y(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4)}{8}$
60	$\mathbb{Q}_{7,1}(E_{1u}, 1)$	E_{1u}	7	Q, T	1	1	$\frac{\sqrt{429}x(x^6-21x^4y^2+35x^2y^4-7y^6)}{32}$
61	$\mathbb{Q}_{7,2}(E_{1u}, 1)$					2	$\frac{\sqrt{429}y(7x^6-35x^4y^2+21x^2y^4-y^6)}{32}$
62	$\mathbb{Q}_{2,1}(E_{2g})$	E_{2g}	2	Q, T	-	1	$\frac{\sqrt{3}(x^2-y^2)}{2}$
63	$\mathbb{Q}_{2,2}(E_{2g})$					2	$-\sqrt{3}xy$
64	$\mathbb{G}_{3,1}(E_{2g})$	E_{2g}	3	G, M	-	1	$\sqrt{15}xyz$
65	$\mathbb{G}_{3,2}(E_{2g})$					2	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
66	$\mathbb{Q}_{4,1}(E_{2g}, 1)$	E_{2g}	4	Q, T	1	1	$\frac{\sqrt{35}(x^4-6x^2y^2+y^4)}{8}$
67	$\mathbb{Q}_{4,2}(E_{2g}, 1)$					2	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
68	$\mathbb{Q}_{4,1}(E_{2g}, 2)$	E_{2g}	4	Q, T	2	1	$\frac{\sqrt{5}(-x^4+6x^2z^2+y^4-6y^2z^2)}{4}$
69	$\mathbb{Q}_{4,2}(E_{2g}, 2)$					2	$\frac{\sqrt{5}xy(x^2+y^2-6z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
70	$\mathbb{Q}_{6,1}(E_{2g}, 1)$	E_{2g}	6	Q, T	1	1	$\frac{3\sqrt{7}(-x^6+5x^4y^2+10x^4z^2+5x^2y^4-60x^2y^2z^2-y^6+10y^4z^2)}{16}$
71	$\mathbb{Q}_{6,2}(E_{2g}, 1)$					2	$\frac{3\sqrt{7}xy(-x^4+10x^2z^2+y^4-10y^2z^2)}{4}$
72	$\mathbb{Q}_{6,1}(E_{2g}, 2)$	E_{2g}	6	Q, T	2	1	$\frac{\sqrt{210}(x^6+x^4y^2-16x^4z^2-x^2y^4+16x^2z^4-y^6+16y^4z^2-16y^2z^4)}{32}$
73	$\mathbb{Q}_{6,2}(E_{2g}, 2)$					2	$\frac{\sqrt{210}xy(-x^4-2x^2y^2+16x^2z^2-y^4+16y^2z^2-16z^4)}{16}$
74	$\mathbb{Q}_{8,1}(E_{2g}, 1)$	E_{2g}	8	Q, T	1	1	$\frac{3\sqrt{715}(x^8-28x^6y^2+70x^4y^4-28x^2y^6+y^8)}{128}$
75	$\mathbb{Q}_{8,2}(E_{2g}, 1)$					2	$\frac{3\sqrt{715}xy(-x^6+7x^4y^2-7x^2y^4+y^6)}{16}$
76	$\mathbb{G}_{2,1}(E_{2u})$	E_{2u}	2	G, M	-	1	$\frac{\sqrt{3}(x^2-y^2)}{2}$
77	$\mathbb{G}_{2,2}(E_{2u})$					2	$-\sqrt{3}xy$
78	$\mathbb{Q}_{3,1}(E_{2u})$	E_{2u}	3	Q, T	-	1	$\sqrt{15}xyz$
79	$\mathbb{Q}_{3,2}(E_{2u})$					2	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
80	$\mathbb{G}_{4,1}(E_{2u}, 1)$	E_{2u}	4	G, M	1	1	$\frac{\sqrt{35}(x^4-6x^2y^2+y^4)}{8}$
81	$\mathbb{G}_{4,2}(E_{2u}, 1)$					2	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
82	$\mathbb{G}_{4,1}(E_{2u}, 2)$	E_{2u}	4	G, M	2	1	$\frac{\sqrt{5}(-x^4+6x^2z^2+y^4-6y^2z^2)}{4}$
83	$\mathbb{G}_{4,2}(E_{2u}, 2)$					2	$\frac{\sqrt{5}xy(x^2+y^2-6z^2)}{2}$
84	$\mathbb{Q}_{5,1}(E_{2u}, 1)$	E_{2u}	5	Q, T	1	1	$\frac{3\sqrt{35}xyz(-x^2+y^2)}{2}$
85	$\mathbb{Q}_{5,2}(E_{2u}, 1)$					2	$\frac{3\sqrt{35}z(x^4-6x^2y^2+y^4)}{8}$
86	$\mathbb{Q}_{5,1}(E_{2u}, 2)$	E_{2u}	5	Q, T	2	1	$\frac{\sqrt{105}xyz(-x^2-y^2+2z^2)}{2}$
87	$\mathbb{Q}_{5,2}(E_{2u}, 2)$					2	$\frac{\sqrt{105}z(-x^4+2x^2z^2+y^4-2y^2z^2)}{4}$
88	$\mathbb{Q}_{9,1}(E_{2u}, 1)$	E_{2u}	9	Q, T	1	1	$\frac{3\sqrt{12155}xyz(x^6-7x^4y^2+7x^2y^4-y^6)}{16}$
89	$\mathbb{Q}_{9,2}(E_{2u}, 1)$					2	$\frac{3\sqrt{12155}z(x^8-28x^6y^2+70x^4y^4-28x^2y^6+y^8)}{128}$

Table 3: dimension = 40

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$ d_v\rangle @\text{Mn}(1)$	1	$ d_{xy}\rangle @\text{Mn}(1)$	2	$ d_{xz}\rangle @\text{Mn}(1)$	3	$ d_{yz}\rangle @\text{Mn}(1)$	4	$ d_u\rangle @\text{Mn}(1)$
5	$ d_v\rangle @\text{Mn}(2)$	6	$ d_{xy}\rangle @\text{Mn}(2)$	7	$ d_{xz}\rangle @\text{Mn}(2)$	8	$ d_{yz}\rangle @\text{Mn}(2)$	9	$ d_u\rangle @\text{Mn}(2)$
10	$ d_v\rangle @\text{Mn}(3)$	11	$ d_{xy}\rangle @\text{Mn}(3)$	12	$ d_{xz}\rangle @\text{Mn}(3)$	13	$ d_{yz}\rangle @\text{Mn}(3)$	14	$ d_u\rangle @\text{Mn}(3)$
15	$ d_v\rangle @\text{Mn}(4)$	16	$ d_{xy}\rangle @\text{Mn}(4)$	17	$ d_{xz}\rangle @\text{Mn}(4)$	18	$ d_{yz}\rangle @\text{Mn}(4)$	19	$ d_u\rangle @\text{Mn}(4)$
20	$ d_v\rangle @\text{Mn}(5)$	21	$ d_{xy}\rangle @\text{Mn}(5)$	22	$ d_{xz}\rangle @\text{Mn}(5)$	23	$ d_{yz}\rangle @\text{Mn}(5)$	24	$ d_u\rangle @\text{Mn}(5)$
25	$ d_v\rangle @\text{Mn}(6)$	26	$ d_{xy}\rangle @\text{Mn}(6)$	27	$ d_{xz}\rangle @\text{Mn}(6)$	28	$ d_{yz}\rangle @\text{Mn}(6)$	29	$ d_u\rangle @\text{Mn}(6)$
30	$ d_v\rangle @\text{Sn}(1)$	31	$ d_{xy}\rangle @\text{Sn}(1)$	32	$ d_{xz}\rangle @\text{Sn}(1)$	33	$ d_{yz}\rangle @\text{Sn}(1)$	34	$ d_u\rangle @\text{Sn}(1)$
35	$ d_v\rangle @\text{Sn}(2)$	36	$ d_{xy}\rangle @\text{Sn}(2)$	37	$ d_{xz}\rangle @\text{Sn}(2)$	38	$ d_{yz}\rangle @\text{Sn}(2)$	39	$ d_u\rangle @\text{Sn}(2)$

Table 4: Atomic basis (orbital part only)

orbital	definition
$ d_v\rangle$	$\frac{\sqrt{3}(x^2-y^2)}{2}$
$ d_{xy}\rangle$	$\sqrt{3}xy$
$ d_{xz}\rangle$	$\sqrt{3}xz$
$ d_{yz}\rangle$	$\sqrt{3}yz$
$ d_u\rangle$	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$

• Mn : 'Mn' site-cluster

* bra: $\langle d_v |, \langle d_{xy} |, \langle d_{xz} |, \langle d_{yz} |, \langle d_u |$

* ket: $|d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$

* wyckoff: 6h

$$\boxed{\text{z1}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z2}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z3}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z4}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z5}} \quad \mathbb{Q}_4^{(c)}(A_{1g}) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z6}} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 2) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z41}} \quad \mathbb{Q}_7^{(c)}(A_{1u}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z42}} \quad \mathbb{G}_2^{(c)}(A_{1u}) = -\frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z43}} \quad \mathbb{G}_4^{(c)}(A_{1u}) = -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z69}} \quad \mathbb{Q}_6^{(c)}(A_{2g}) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z70}} \quad \mathbb{G}_1^{(c)}(A_{2g}) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z71}} \quad \mathbb{G}_3^{(c)}(A_{2g}) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z95}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z96}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, b) = \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z97}} \quad \mathbb{Q}_3^{(c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z121}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z122}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, b) = \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z123}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, c) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z149}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, a) = \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z150}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, b) = -\mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z151}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, c) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z152}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, d) = \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z153}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, e) = \frac{\sqrt{58} \left(-2\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) + 2\sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{58}$$

$$\boxed{\text{z154}} \quad \mathbb{Q}_5^{(c)}(B_{1u}) = \frac{\sqrt{58} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) + 2\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) - 2\sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{58}$$

$$\boxed{\text{z184}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z185}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, b) = \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z186}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, c) = -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{2}$$

$$\begin{aligned}
\text{z215} \quad \mathbb{Q}_3^{(c)}(B_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{2} \\
\text{z216} \quad \mathbb{Q}_3^{(c)}(B_{2u}, b) &= \frac{\sqrt{58} \left(2\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) + 2\sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{58} \\
\text{z217} \quad \mathbb{Q}_5^{(c)}(B_{2u}) &= \frac{\sqrt{58} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) + 2\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) + 2\sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{58} \\
\text{z241} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z242} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z243} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, b) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} \\
\text{z244} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} \\
\text{z245} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, c) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \sqrt{7} \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \sqrt{7} \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{8} \\
\text{z246} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, c) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) - \sqrt{7} \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \sqrt{7} \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{8} \\
\text{z247} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z248} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z249} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, b) &= \frac{\sqrt{2} \left(-\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{8} \\
\text{z250} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, b) &= \frac{\sqrt{2} \left(\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) - \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{8}
\end{aligned}$$

$$\boxed{\text{z251}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1) = \frac{-\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2}$$

$$\boxed{\text{z252}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1) = \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}$$

$$\boxed{\text{z359}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2}$$

$$\boxed{\text{z360}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2}$$

$$\boxed{\text{z361}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, b) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$$\boxed{\text{z362}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$$\boxed{\text{z363}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, c) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z364}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, c) = -\frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z365}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, d) = \frac{\sqrt{58} \left(2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(s)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z366}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, d) = -\frac{\sqrt{58} \left(2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(s)}(B_{1u}) + \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z367}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, a) = \frac{\sqrt{7} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z368}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, a) = \frac{\sqrt{7} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z369}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, b) = \frac{\sqrt{58} \left(\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(s)}(B_{1u}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58}$$

$$\begin{aligned}
\text{z370} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, b) &= \frac{\sqrt{58} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_3^{(s)}(B_{1u}) + 2\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58} \\
\text{z371} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, c) &= \frac{\sqrt{7} \left(\sqrt{5} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \sqrt{5} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) - 2\mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{14} \\
\text{z372} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, c) &= -\frac{\sqrt{7} \left(\sqrt{5} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \sqrt{5} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) + 2\mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{14} \\
\text{z373} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2} \\
\text{z374} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1) &= -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2} \\
\text{z375} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 2) &= \frac{\sqrt{14} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \sqrt{5} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{14} \\
\text{z376} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 2) &= \frac{\sqrt{14} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \sqrt{5} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{14} \\
\text{z481} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} \\
\text{z482} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} \\
\text{z483} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, b) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z484} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, b) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z485} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, c) &= -\frac{\sqrt{2} \mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} \\
\text{z486} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, c) &= -\frac{\sqrt{2} \mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2}
\end{aligned}$$

$$\begin{aligned}
\text{z487} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, d) &= \frac{\sqrt{142} \left(\sqrt{35} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) - \sqrt{35} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{142} \\
\text{z488} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, d) &= \frac{\sqrt{142} \left(-\sqrt{35} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) - \sqrt{35} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{142} \\
\text{z489} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1a) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} \\
\text{z490} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1a) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} \\
\text{z491} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1b) &= \frac{\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z492} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1b) &= \frac{\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z493} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1c) &= \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(s)}(E_{2g})}{2} \\
\text{z494} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1c) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(s)}(E_{2g})}{2} \\
\text{z495} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2a) &= \frac{\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z496} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2a) &= \frac{\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\text{z497} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2b) &= \frac{\sqrt{71} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) - 2\sqrt{35} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) \right)}{142} \\
\text{z498} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2b) &= -\frac{\sqrt{71} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(s)}(E_{2g}) + 2\sqrt{35} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(s)}(E_{2g}) \right)}{142} \\
\text{z613} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, a) &= -\frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_3^{(s)}(B_{1u})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z614}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(s)}(B_{1u})}{2} \\
\boxed{\text{z615}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2} \\
\boxed{\text{z616}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, b) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2} \\
\boxed{\text{z617}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, c) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(s)}(B_{1u})}{2} \\
\boxed{\text{z618}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, c) &= -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(s)}(B_{1u})}{2} \\
\boxed{\text{z619}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, d) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{8} \\
\boxed{\text{z620}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, d) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{8} \\
\boxed{\text{z621}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1) &= -\frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u})}{2} \\
\boxed{\text{z622}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1) &= \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u})}{2} \\
\boxed{\text{z623}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 2) &= \frac{\sqrt{2} \left(\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) \right)}{8} \\
\boxed{\text{z624}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 2) &= \frac{\sqrt{2} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(s)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(s)}(E_{1u}) \right)}{8}
\end{aligned}$$

• **Sn** : 'Sn' site-cluster

- * bra: $\langle d_v |, \langle d_{xy} |, \langle d_{xz} |, \langle d_{yz} |, \langle d_u |$
- * ket: $|d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$
- * wyckoff: **2c**

$$\boxed{\text{z7}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z8}} \quad \mathbb{Q}_2^{(c)}(A_{1g}) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_4^{(c)}(A_{1g}) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z44}} \quad \mathbb{Q}_7^{(c)}(A_{1u}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z98}} \quad \mathbb{Q}_1^{(c)}(A_{2u}) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z124}} \quad \mathbb{Q}_4^{(c)}(B_{1g}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z155}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z156}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z157}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, c) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$$

$$\boxed{\text{z187}} \quad \mathbb{Q}_4^{(c)}(B_{2g}) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z253}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z254}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z255}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z256}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z377}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, a) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$$\boxed{\text{z378}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$$\boxed{\text{z379}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, b) = \frac{\sqrt{58} \left(2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(s)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z380}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, b) = -\frac{\sqrt{58} \left(2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(s)}(B_{1u}) + \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z381}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}) = \frac{\sqrt{58} \left(\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(s)}(B_{1u}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z382}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}) = \frac{\sqrt{58} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(s)}(B_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(s)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z499}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z500}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z501}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z502}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z503}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z504}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z625}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, a) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$$\boxed{\text{z626}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$$\boxed{\text{z627}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, b) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

$$\boxed{\text{z628}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, b) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(s)}(B_{1u})}{2}$$

• **Mn;Mn_001_1** : 'Mn'-'Mn' bond-cluster

* bra: $\langle d_v |, \langle d_{xy} |, \langle d_{xz} |, \langle d_{yz} |, \langle d_u |$

* ket: $|d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$

* wyckoff: **6b@6h**

$$\boxed{\text{z10}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z11}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z12}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, c) = \mathbb{M}_1^{(a)}(A_{2g})\mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\text{z13}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z14}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z15}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, c) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\text{z16}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, d) = -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z17}} \quad \mathbb{Q}_4^{(c)}(A_{1g}) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z18}} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 2) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z45}} \quad \mathbb{Q}_7^{(c)}(A_{1u}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z46}} \quad \mathbb{G}_0^{(c)}(A_{1u}, a) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z47}} \quad \mathbb{G}_0^{(c)}(A_{1u}, b) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_3^{(b)}(B_{2u})$$

$$\boxed{\text{z48}} \quad \mathbb{G}_2^{(c)}(A_{1u}, a) = -\frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z49}} \quad \mathbb{G}_2^{(c)}(A_{1u}, b) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{2}$$

$$\begin{aligned}
\boxed{\text{z50}} \quad \mathbb{G}_4^{(c)}(A_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{2} \\
\boxed{\text{z72}} \quad \mathbb{Q}_6^{(c)}(A_{2g}) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2} \\
\boxed{\text{z73}} \quad \mathbb{G}_1^{(c)}(A_{2g}, a) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2} \\
\boxed{\text{z74}} \quad \mathbb{G}_1^{(c)}(A_{2g}, b) &= \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) \right)}{2} \\
\boxed{\text{z75}} \quad \mathbb{G}_3^{(c)}(A_{2g}) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2} \\
\boxed{\text{z99}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{2} \\
\boxed{\text{z100}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, b) &= \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_3^{(b)}(B_{1u}) \\
\boxed{\text{z101}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, c) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{2} \\
\boxed{\text{z102}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, d) &= -\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_3^{(b)}(B_{2u}) \\
\boxed{\text{z103}} \quad \mathbb{Q}_3^{(c)}(A_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{2} \\
\boxed{\text{z104}} \quad \mathbb{Q}_3^{(c)}(A_{2u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{2} \\
\boxed{\text{z125}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2} \\
\boxed{\text{z126}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, b) &= \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z127}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, c) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2}
\end{aligned}$$

$$\boxed{\text{z128}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, d) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\text{z129}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, e) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) - \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z130}} \quad \mathbb{G}_3^{(c)}(B_{1g}) = \frac{\sqrt{2} \left(-\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z158}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z159}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z160}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, c) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z161}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, d) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z162}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, e) = \frac{\sqrt{58} \left(-2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{58}$$

$$\boxed{\text{z163}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, f) = \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u})$$

$$\boxed{\text{z164}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, g) = -\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u})$$

$$\boxed{\text{z165}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, h) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z166}} \quad \mathbb{Q}_5^{(c)}(B_{1u}) = \frac{\sqrt{58} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{58}$$

$$\boxed{\text{z188}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z189}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, b) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z190}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, c) = -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z191}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, d) = \mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\text{z192}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, e) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z193}} \quad \mathbb{G}_3^{(c)}(B_{2g}) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z218}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z219}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, b) = \frac{\sqrt{58} \left(2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{58}$$

$$\boxed{\text{z220}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, c) = -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z221}} \quad \mathbb{Q}_5^{(c)}(B_{2u}) = \frac{\sqrt{58} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{58}$$

$$\boxed{\text{z257}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z258}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z259}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, b) = \frac{-\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z260}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z261}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, c) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{8}$$

$$\boxed{\text{z262}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, c) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{8}$$

$$\begin{aligned}
\boxed{\text{z263}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, d) &= \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z264}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, d) &= -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z265}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, e) &= \frac{-\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z266}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, e) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z267}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, f) &= \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z268}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, f) &= -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z269}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, g) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) \right)}{8} \\
\boxed{\text{z270}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, g) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) \right)}{8} \\
\boxed{\text{z271}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z272}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z273}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, b) &= \frac{\sqrt{2} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{8} \\
\boxed{\text{z274}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, b) &= \frac{\sqrt{2} \left(\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{8} \\
\boxed{\text{z275}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, c) &= \frac{\sqrt{2} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) \right)}{8}
\end{aligned}$$

$$\boxed{\text{z276}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, c) = \frac{\sqrt{2} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) \right)}{8}$$

$$\boxed{\text{z277}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1) = \frac{-\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z278}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1) = \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z279}} \quad \mathbb{G}_{5,1}^{(c)}(E_{1g}, 1) = \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z280}} \quad \mathbb{G}_{5,2}^{(c)}(E_{1g}, 1) = \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z383}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z384}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z385}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, b) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z386}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z387}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, c) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z388}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, c) = -\frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z389}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, d) = \frac{\sqrt{58} \left(2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z390}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, d) = -\frac{\sqrt{58} \left(2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) \right)}{58}$$

$$\begin{aligned}
\text{z391} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, e) &= -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} \\
\text{z392} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, e) &= \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} \\
\text{z393} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, f) &= \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{2u})}{2} \\
\text{z394} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, f) &= \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{2u})}{2} \\
\text{z395} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, a) &= \frac{\sqrt{7} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14} \\
\text{z396} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, a) &= \frac{\sqrt{7} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14} \\
\text{z397} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, b) &= \frac{\sqrt{58} \left(\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) \right)}{58} \\
\text{z398} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, b) &= \frac{\sqrt{58} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) \right)}{58} \\
\text{z399} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, c) &= \frac{\sqrt{7} \left(\sqrt{5}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \sqrt{5}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - 2\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14} \\
\text{z400} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, c) &= -\frac{\sqrt{7} \left(\sqrt{5}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14} \\
\text{z401} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, d) &= \frac{\sqrt{11} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - 2\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{22} \\
\text{z402} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, d) &= \frac{\sqrt{11} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + 2\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{22} \\
\text{z403} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}
\end{aligned}$$

$$\boxed{\text{z404}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z405}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 2) = \frac{\sqrt{14} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z406}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 2) = \frac{\sqrt{14} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z407}} \quad \mathbb{G}_{2,1}^{(c)}(E_{1u}) = \frac{\sqrt{22} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \sqrt{5}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{22}$$

$$\boxed{\text{z408}} \quad \mathbb{G}_{2,2}^{(c)}(E_{1u}) = \frac{\sqrt{22} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{22}$$

$$\boxed{\text{z505}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z506}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z507}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z508}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z509}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, c) = -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z510}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, c) = -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z511}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, d) = \frac{\sqrt{142} \left(\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{142}$$

$$\boxed{\text{z512}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, d) = \frac{\sqrt{142} \left(-\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{142}$$

$$\begin{aligned}
\boxed{\text{z513}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, e) &= \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z514}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, e) &= -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z515}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, f) &= \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z516}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, f) &= -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z517}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, g) &= -\frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z518}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, g) &= \frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z519}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1a) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z520}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1a) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z521}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1b) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z522}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1b) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z523}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1c) &= \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z524}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1c) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z525}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1d) &= \frac{-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z526}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1d) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g})}{2}
\end{aligned}$$

$$\boxed{\text{z527}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z528}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z529}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2b) = \frac{\sqrt{71} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2\sqrt{35}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{142}$$

$$\boxed{\text{z530}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2b) = -\frac{\sqrt{71} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 2\sqrt{35}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{142}$$

$$\boxed{\text{z629}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, a) = -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z630}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z631}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z632}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, b) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z633}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, c) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z634}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, c) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z635}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, d) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z636}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, d) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z637}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, e) = -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2}$$

$$\boxed{\text{z638}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, e) = -\frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2}$$

$$\boxed{\text{z639}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, f) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2}$$

$$\boxed{\text{z640}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, f) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2}$$

$$\boxed{\text{z641}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, g) = \frac{\sqrt{2} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z642}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, g) = \frac{\sqrt{2} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z643}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1) = -\frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z644}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1) = \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z645}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 2) = \frac{\sqrt{2} \left(\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z646}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 2) = \frac{\sqrt{2} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z647}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, a) = \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z648}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, a) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z649}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, b) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z650}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, b) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{8}$$

$$\boxed{\text{z651}} \quad \mathbb{G}_{4,1}^{(c)}(E_{2u}, 1) = \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z652}} \quad \mathbb{G}_{4,2}^{(c)}(E_{2u}, 1) = \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2}$$

- **Sn;Mn_001_1** : 'Mn'-'Sn' bond-cluster

* bra: $\langle d_v |$, $\langle d_{xy} |$, $\langle d_{xz} |$, $\langle d_{yz} |$, $\langle d_u |$

* ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$

* wyckoff: **12a@12j**

$$\boxed{\text{z19}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z20}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z21}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, c) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) \right)}{2}$$

$$\boxed{\text{z22}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, d) = \mathbb{M}_1^{(a)}(A_{2g})\mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\text{z23}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z24}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z25}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, c) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\text{z26}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, d) = -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) \right)}{2}$$

$$\boxed{\text{z27}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, e) = -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) \right)}{2}$$

$$\boxed{\text{z28}} \quad \mathbb{Q}_4^{(c)}(A_{1g}) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z29}} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 2a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) \right)}{2}$$

$$\begin{aligned}
\text{z30} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 2b) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2} \\
\text{z31} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 2c) &= -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) \right)}{2} \\
\text{z51} \quad \mathbb{Q}_7^{(c)}(A_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) \right)}{2} \\
\text{z52} \quad \mathbb{Q}_7^{(c)}(A_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_3^{(b)}(B_{2u}) \right)}{2} \\
\text{z53} \quad \mathbb{Q}_7^{(c)}(A_{1u}, c) &= -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) \right)}{2} \\
\text{z54} \quad \mathbb{G}_0^{(c)}(A_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) \right)}{2} \\
\text{z55} \quad \mathbb{G}_0^{(c)}(A_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) \right)}{2} \\
\text{z56} \quad \mathbb{G}_0^{(c)}(A_{1u}, c) &= \frac{\sqrt{2} \left(\mathbb{M}_3^{(a)}(B_{1g}) \mathbb{T}_3^{(b)}(B_{1u}) + \mathbb{M}_3^{(a)}(B_{2g}) \mathbb{T}_3^{(b)}(B_{2u}) \right)}{2} \\
\text{z57} \quad \mathbb{G}_2^{(c)}(A_{1u}, a) &= -\frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{2} \\
\text{z58} \quad \mathbb{G}_2^{(c)}(A_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_3^{(b)}(B_{1u}) - \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_3^{(b)}(B_{2u}) \right)}{2} \\
\text{z59} \quad \mathbb{G}_2^{(c)}(A_{1u}, c) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) \right)}{2} \\
\text{z60} \quad \mathbb{G}_2^{(c)}(A_{1u}, d) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) \right)}{2} \\
\text{z61} \quad \mathbb{G}_4^{(c)}(A_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{2}
\end{aligned}$$

$$\boxed{\text{z62}} \quad \mathbb{G}_6^{(c)}(A_{1u}, 2) = \frac{\sqrt{2} \left(-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_3^{(b)}(B_{1u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_3^{(b)}(B_{2u}) \right)}{2}$$

$$\boxed{\text{z76}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_6^{(b)}(A_{2g})$$

$$\boxed{\text{z77}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_6^{(b)}(A_{2g})$$

$$\boxed{\text{z78}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, c) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) \right)}{2}$$

$$\boxed{\text{z79}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, d) = \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_6^{(b)}(A_{2g})$$

$$\boxed{\text{z80}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, e) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z81}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, f) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) \right)}{2}$$

$$\boxed{\text{z82}} \quad \mathbb{G}_1^{(c)}(A_{2g}, a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z83}} \quad \mathbb{G}_1^{(c)}(A_{2g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) \right)}{2}$$

$$\boxed{\text{z84}} \quad \mathbb{G}_1^{(c)}(A_{2g}, c) = \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z85}} \quad \mathbb{G}_1^{(c)}(A_{2g}, d) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) \right)}{2}$$

$$\boxed{\text{z86}} \quad \mathbb{G}_1^{(c)}(A_{2g}, e) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) \right)}{2}$$

$$\boxed{\text{z87}} \quad \mathbb{G}_3^{(c)}(A_{2g}, a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2}$$

$$\boxed{\text{z88}} \quad \mathbb{G}_3^{(c)}(A_{2g}, b) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_0^{(b)}(A_{1g})$$

$$\begin{aligned}
\text{z105} \quad \mathbb{Q}_1^{(c)}(A_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{2} \\
\text{z106} \quad \mathbb{Q}_1^{(c)}(A_{2u}, b) &= \frac{\sqrt{2} \left(\mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_3^{(b)}(B_{2u}) + \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_3^{(b)}(B_{1u}) \right)}{2} \\
\text{z107} \quad \mathbb{Q}_1^{(c)}(A_{2u}, c) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) - \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) \right)}{2} \\
\text{z108} \quad \mathbb{Q}_1^{(c)}(A_{2u}, d) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) - \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) \right)}{2} \\
\text{z109} \quad \mathbb{Q}_1^{(c)}(A_{2u}, e) &= \frac{\sqrt{2} \left(-\mathbb{M}_3^{(a)}(B_{1g}) \mathbb{T}_3^{(b)}(B_{2u}) + \mathbb{M}_3^{(a)}(B_{2g}) \mathbb{T}_3^{(b)}(B_{1u}) \right)}{2} \\
\text{z110} \quad \mathbb{Q}_3^{(c)}(A_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{2} \\
\text{z111} \quad \mathbb{Q}_3^{(c)}(A_{2u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) - \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) \right)}{2} \\
\text{z112} \quad \mathbb{Q}_3^{(c)}(A_{2u}, c) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) - \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) \right)}{2} \\
\text{z113} \quad \mathbb{Q}_7^{(c)}(A_{2u}, 2a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) \right)}{2} \\
\text{z114} \quad \mathbb{Q}_7^{(c)}(A_{2u}, 2b) &= \frac{\sqrt{2} \left(\mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_3^{(b)}(B_{2u}) - \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_3^{(b)}(B_{1u}) \right)}{2} \\
\text{z115} \quad \mathbb{Q}_7^{(c)}(A_{2u}, 2c) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) \right)}{2} \\
\text{z116} \quad \mathbb{G}_6^{(c)}(A_{2u}) &= \frac{\sqrt{2} \left(\mathbb{M}_3^{(a)}(B_{1g}) \mathbb{T}_3^{(b)}(B_{2u}) + \mathbb{M}_3^{(a)}(B_{2g}) \mathbb{T}_3^{(b)}(B_{1u}) \right)}{2} \\
\text{z131} \quad \mathbb{Q}_4^{(c)}(B_{1g}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z132}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, b) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) \right)}{2} \\
\boxed{\text{z133}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, c) &= \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z134}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, d) &= \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_6^{(b)}(A_{2g}) \\
\boxed{\text{z135}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, e) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2} \\
\boxed{\text{z136}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, f) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) \right)}{2} \\
\boxed{\text{z137}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, g) &= \mathbb{M}_3^{(a)}(B_{2g}) \mathbb{M}_1^{(b)}(A_{2g}) \\
\boxed{\text{z138}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, h) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, a) - \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, a) \right)}{2} \\
\boxed{\text{z139}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, i) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, b) - \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, b) \right)}{2} \\
\boxed{\text{z140}} \quad \mathbb{G}_3^{(c)}(B_{1g}, a) &= \frac{\sqrt{2} \left(-\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, a) \right)}{2} \\
\boxed{\text{z141}} \quad \mathbb{G}_3^{(c)}(B_{1g}, b) &= \frac{\sqrt{2} \left(-\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, b) \right)}{2} \\
\boxed{\text{z142}} \quad \mathbb{G}_3^{(c)}(B_{1g}, c) &= \mathbb{M}_3^{(a)}(B_{1g}) \mathbb{T}_0^{(b)}(A_{1g}) \\
\boxed{\text{z167}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, a) &= \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_3^{(b)}(B_{1u}) \\
\boxed{\text{z168}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, b) &= -\mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_3^{(b)}(B_{1u}) \\
\boxed{\text{z169}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, c) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{2} \\
\boxed{\text{z170}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, d) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z171}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, e) &= \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u}) \\
\boxed{\text{z172}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, f) &= \frac{\sqrt{58} \left(-2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{58} \\
\boxed{\text{z173}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, g) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) \right)}{2} \\
\boxed{\text{z174}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, h) &= \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u}) \\
\boxed{\text{z175}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, i) &= -\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{2u}) \\
\boxed{\text{z176}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, j) &= \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) \right)}{2} \\
\boxed{\text{z177}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, k) &= \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) \right)}{2} \\
\boxed{\text{z178}} \quad \mathbb{Q}_5^{(c)}(B_{1u}) &= \frac{\sqrt{58} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{58} \\
\boxed{\text{z179}} \quad \mathbb{Q}_9^{(c)}(B_{1u}, 1) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) \right)}{2} \\
\boxed{\text{z194}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2} \\
\boxed{\text{z195}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, b) &= -\frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) \right)}{2} \\
\boxed{\text{z196}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, c) &= \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z197}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, d) &= \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_6^{(b)}(A_{2g}) \\
\boxed{\text{z198}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, e) &= -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2}
\end{aligned}$$

$$\boxed{\text{z199}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, f) = \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) \right)}{2}$$

$$\boxed{\text{z200}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, g) = \mathbb{M}_3^{(a)}(B_{1g}) \mathbb{M}_1^{(b)}(A_{2g})$$

$$\boxed{\text{z201}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, h) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, a) \right)}{2}$$

$$\boxed{\text{z202}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, i) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, b) \right)}{2}$$

$$\boxed{\text{z203}} \quad \mathbb{G}_3^{(c)}(B_{2g}, a) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, a) \right)}{2}$$

$$\boxed{\text{z204}} \quad \mathbb{G}_3^{(c)}(B_{2g}, b) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{2,2}^{(b)}(E_{2g}, b) \right)}{2}$$

$$\boxed{\text{z205}} \quad \mathbb{G}_3^{(c)}(B_{2g}, c) = \mathbb{M}_3^{(a)}(B_{2g}) \mathbb{T}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z222}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, a) = \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_3^{(b)}(B_{2u})$$

$$\boxed{\text{z223}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, b) = -\mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_3^{(b)}(B_{2u})$$

$$\boxed{\text{z224}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, c) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z225}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, d) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) \right)}{2}$$

$$\boxed{\text{z226}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, e) = \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_3^{(b)}(B_{2u})$$

$$\boxed{\text{z227}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, f) = \frac{\sqrt{58} \left(2\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{58}$$

$$\boxed{\text{z228}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, g) = -\frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) \right)}{2}$$

$$\boxed{\text{z229}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, h) = -\mathbb{M}_1^{(a)}(A_{2g}) \mathbb{T}_3^{(b)}(B_{1u})$$

$$\begin{aligned}
\boxed{\text{z230}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, i) &= \mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{1u}) \\
\boxed{\text{z231}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, j) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) \right)}{2} \\
\boxed{\text{z232}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, k) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) \right)}{2} \\
\boxed{\text{z233}} \quad \mathbb{Q}_5^{(c)}(B_{2u}) &= \frac{\sqrt{58} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{58} \\
\boxed{\text{z234}} \quad \mathbb{Q}_9^{(c)}(B_{2u}, 1) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) \right)}{2} \\
\boxed{\text{z281}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z282}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z283}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, b) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z284}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z285}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, c) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{8} \\
\boxed{\text{z286}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, c) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{8} \\
\boxed{\text{z287}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, d) &= \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) - \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z288}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, d) &= -\frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2}
\end{aligned}$$

$$\begin{aligned}
\text{z289} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, e) &= \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\text{z290} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, e) &= -\frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\text{z291} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, f) &= \frac{-\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a)}{2} \\
\text{z292} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, f) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a)}{2} \\
\text{z293} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, g) &= \frac{-\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b)}{2} \\
\text{z294} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, g) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b)}{2} \\
\text{z295} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, h) &= \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\text{z296} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, h) &= -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\text{z297} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, i) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) \right)}{8} \\
\text{z298} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, i) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) \right)}{8} \\
\text{z299} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, j) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) \right)}{8} \\
\text{z300} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, j) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) \right)}{8} \\
\text{z301} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}
\end{aligned}$$

$$\begin{aligned}
\text{z302} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, a) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\text{z303} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, b) &= \frac{\sqrt{2} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{8} \\
\text{z304} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, b) &= \frac{\sqrt{2} \left(\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{8} \\
\text{z305} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, c) &= \frac{\sqrt{2} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) \right)}{8} \\
\text{z306} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, c) &= \frac{\sqrt{2} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) \right)}{8} \\
\text{z307} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, d) &= \frac{\sqrt{2} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) \right)}{8} \\
\text{z308} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, d) &= \frac{\sqrt{2} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) \right)}{8} \\
\text{z309} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1a) &= -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_6^{(b)}(A_{2g})}{2} \\
\text{z310} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_6^{(b)}(A_{2g})}{2} \\
\text{z311} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} \\
\text{z312} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1b) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} \\
\text{z313} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1c) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_6^{(b)}(A_{2g})}{2} \\
\text{z314} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1c) &= -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_6^{(b)}(A_{2g})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z315}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1d) &= \frac{-\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z316}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1d) &= \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z317}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1e) &= \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z318}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1e) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z319}} \quad \mathbb{Q}_{8,1}^{(c)}(E_{1g}, 1) &= \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z320}} \quad \mathbb{Q}_{8,2}^{(c)}(E_{1g}, 1) &= \frac{-\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z321}} \quad \mathbb{G}_{1,1}^{(c)}(E_{1g}) &= \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z322}} \quad \mathbb{G}_{1,2}^{(c)}(E_{1g}) &= \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z323}} \quad \mathbb{G}_{3,1}^{(c)}(E_{1g}) &= \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z324}} \quad \mathbb{G}_{3,2}^{(c)}(E_{1g}) &= \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z325}} \quad \mathbb{G}_{5,1}^{(c)}(E_{1g}, 1a) &= \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a)}{2} \\
\boxed{\text{z326}} \quad \mathbb{G}_{5,2}^{(c)}(E_{1g}, 1a) &= \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a)}{2} \\
\boxed{\text{z327}} \quad \mathbb{G}_{5,1}^{(c)}(E_{1g}, 1b) &= \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b)}{2} \\
\boxed{\text{z328}} \quad \mathbb{G}_{5,2}^{(c)}(E_{1g}, 1b) &= \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b)}{2}
\end{aligned}$$

$$\boxed{\text{z409}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z410}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z411}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{2u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z412}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{2u})}{2}$$

$$\boxed{\text{z413}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, c) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z414}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, c) = -\frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14}$$

$$\boxed{\text{z415}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, d) = \frac{\sqrt{29} \left(2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{2u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{2u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) \right)}{58}$$

$$\boxed{\text{z416}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, d) = \frac{\sqrt{29} \left(-2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) - \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{2u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{2u}) \right)}{58}$$

$$\boxed{\text{z417}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, e) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2}$$

$$\boxed{\text{z418}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, e) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2}$$

$$\boxed{\text{z419}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, f) = -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a)}{2}$$

$$\boxed{\text{z420}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, f) = \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a)}{2}$$

$$\boxed{\text{z421}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, g) = -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b)}{2}$$

$$\begin{aligned}
\boxed{\text{z422}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, g) &= \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b)}{2} \\
\boxed{\text{z423}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, h) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{2u}) - \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z424}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, h) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{1u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{2u})}{2} \\
\boxed{\text{z425}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, a) &= \frac{\sqrt{7} \left(\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14} \\
\boxed{\text{z426}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, a) &= \frac{\sqrt{7} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14} \\
\boxed{\text{z427}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, b) &= \frac{\sqrt{29} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{2u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{2u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) \right)}{58} \\
\boxed{\text{z428}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, b) &= \frac{\sqrt{29} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{1u}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_3^{(b)}(B_{2u}) + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{2u}) \right)}{58} \\
\boxed{\text{z429}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, c) &= \frac{\sqrt{7} \left(\sqrt{5}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \sqrt{5}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - 2\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14} \\
\boxed{\text{z430}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, c) &= -\frac{\sqrt{7} \left(\sqrt{5}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{5}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + 2\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14} \\
\boxed{\text{z431}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, d) &= \frac{\sqrt{11} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) - 2\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) \right)}{22} \\
\boxed{\text{z432}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, d) &= \frac{\sqrt{11} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + 2\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) \right)}{22} \\
\boxed{\text{z433}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, e) &= \frac{\sqrt{11} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) - 2\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) \right)}{22} \\
\boxed{\text{z434}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, e) &= \frac{\sqrt{11} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + 2\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) \right)}{22}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z435}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1a) &= \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z436}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1a) &= \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z437}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{2u}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z438}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1b) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_3^{(b)}(B_{2u})}{2} \\
\boxed{\text{z439}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1c) &= -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z440}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1c) &= -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z441}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1d) &= -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{2u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z442}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1d) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_3^{(b)}(B_{2u})}{2} \\
\boxed{\text{z443}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1e) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} \\
\boxed{\text{z444}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1e) &= -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} \\
\boxed{\text{z445}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1f) &= \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z446}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1f) &= \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z447}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1g) &= -\frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z448}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1g) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{1u}) - \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{2u})}{2}
\end{aligned}$$

$$\begin{aligned}
\text{z449} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 2) &= \frac{\sqrt{14} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{5} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{14} \\
\text{z450} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 2) &= \frac{\sqrt{14} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \sqrt{5} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{14} \\
\text{z451} \quad \mathbb{Q}_{7,1}^{(c)}(E_{1u}, 1a) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) - \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\text{z452} \quad \mathbb{Q}_{7,2}^{(c)}(E_{1u}, 1a) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\text{z453} \quad \mathbb{Q}_{7,1}^{(c)}(E_{1u}, 1b) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_3^{(b)}(B_{2u}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_3^{(b)}(B_{1u})}{2} \\
\text{z454} \quad \mathbb{Q}_{7,2}^{(c)}(E_{1u}, 1b) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_3^{(b)}(B_{2u})}{2} \\
\text{z455} \quad \mathbb{Q}_{7,1}^{(c)}(E_{1u}, 1c) &= \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\text{z456} \quad \mathbb{Q}_{7,2}^{(c)}(E_{1u}, 1c) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\text{z457} \quad \mathbb{G}_{2,1}^{(c)}(E_{1u}, a) &= \frac{\sqrt{22} \left(\sqrt{3} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) - \sqrt{3} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) - \sqrt{5} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) \right)}{22} \\
\text{z458} \quad \mathbb{G}_{2,2}^{(c)}(E_{1u}, a) &= \frac{\sqrt{22} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) - \sqrt{3} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \sqrt{5} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) \right)}{22} \\
\text{z459} \quad \mathbb{G}_{2,1}^{(c)}(E_{1u}, b) &= \frac{\sqrt{22} \left(\sqrt{3} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) - \sqrt{3} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) - \sqrt{5} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) \right)}{22} \\
\text{z460} \quad \mathbb{G}_{2,2}^{(c)}(E_{1u}, b) &= \frac{\sqrt{22} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) - \sqrt{3} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \sqrt{5} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) \right)}{22} \\
\text{z531} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}
\end{aligned}$$

$$\boxed{\text{z532}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z533}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z534}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z535}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, c) = -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z536}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, c) = -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z537}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, d) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2}$$

$$\boxed{\text{z538}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, d) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2}$$

$$\boxed{\text{z539}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, e) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_6^{(b)}(A_{2g})}{2}$$

$$\boxed{\text{z540}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, e) = -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_6^{(b)}(A_{2g})}{2}$$

$$\boxed{\text{z541}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, f) = \frac{\sqrt{142} \left(\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{142}$$

$$\boxed{\text{z542}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, f) = \frac{\sqrt{142} \left(-\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{35}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{142}$$

$$\boxed{\text{z543}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, g) = \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2}$$

$$\boxed{\text{z544}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, g) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2}$$

$$\begin{aligned}
\boxed{\text{z545}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, h) &= \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a)}{2} \\
\boxed{\text{z546}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, h) &= -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a)}{2} \\
\boxed{\text{z547}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, i) &= \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b)}{2} \\
\boxed{\text{z548}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, i) &= -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b)}{2} \\
\boxed{\text{z549}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, j) &= \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z550}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, j) &= -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_1^{(b)}(A_{2g})}{2} \\
\boxed{\text{z551}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, k) &= -\frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a)}{2} \\
\boxed{\text{z552}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, k) &= \frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a)}{2} \\
\boxed{\text{z553}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, l) &= -\frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b)}{2} \\
\boxed{\text{z554}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, l) &= \frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b)}{2} \\
\boxed{\text{z555}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1a) &= \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z556}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1a) &= \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z557}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1b) &= -\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_6^{(b)}(A_{2g})}{2} \\
\boxed{\text{z558}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1b) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_6^{(b)}(A_{2g})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z559}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1c) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z560}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1c) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z561}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1d) &= -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z562}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1d) &= -\frac{\sqrt{2}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z563}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1e) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z564}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1e) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z565}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1f) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_6^{(b)}(A_{2g})}{2} \\
\boxed{\text{z566}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1f) &= -\frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_6^{(b)}(A_{2g})}{2} \\
\boxed{\text{z567}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1g) &= \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z568}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1g) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z569}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1h) &= \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z570}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1h) &= \frac{\sqrt{2}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2} \\
\boxed{\text{z571}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1i) &= \frac{-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a)}{2} \\
\boxed{\text{z572}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1i) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, a) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, a)}{2}
\end{aligned}$$

$$\boxed{\text{z573}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1j) = \frac{-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b)}{2}$$

$$\boxed{\text{z574}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1j) = \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{2,2}^{(b)}(E_{2g}, b) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{2,1}^{(b)}(E_{2g}, b)}{2}$$

$$\boxed{\text{z575}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z576}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z577}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2b) = \frac{\sqrt{71} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2\sqrt{35}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{142}$$

$$\boxed{\text{z578}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2b) = -\frac{\sqrt{71} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 2\sqrt{35}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{142}$$

$$\boxed{\text{z579}} \quad \mathbb{Q}_{8,1}^{(c)}(E_{2g}, 1) = \frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1)}{2}$$

$$\boxed{\text{z580}} \quad \mathbb{Q}_{8,2}^{(c)}(E_{2g}, 1) = -\frac{\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1)}{2}$$

$$\boxed{\text{z581}} \quad \mathbb{G}_{3,1}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z582}} \quad \mathbb{G}_{3,2}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z653}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2}$$

$$\boxed{\text{z654}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u})}{2}$$

$$\boxed{\text{z655}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, b) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$$\begin{aligned}
\text{z656} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, b) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} \\
\text{z657} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, c) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2} \\
\text{z658} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, c) &= \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u})}{2} \\
\text{z659} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, d) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{8} \\
\text{z660} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, d) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{8} \\
\text{z661} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, e) &= -\frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\text{z662} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, e) &= \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) - \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\text{z663} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, f) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\text{z664} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, f) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2} \\
\text{z665} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, g) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\text{z666} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, g) &= \frac{-\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2} \\
\text{z667} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, h) &= \frac{\sqrt{2} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) \right)}{8} \\
\text{z668} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, h) &= \frac{\sqrt{2} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) \right)}{8}
\end{aligned}$$

$$\begin{aligned}
\text{z669} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, i) &= \frac{\sqrt{2} \left(\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) \right)}{8} \\
\text{z670} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, i) &= \frac{\sqrt{2} \left(-\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) \right)}{8} \\
\text{z671} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1a) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2} \\
\text{z672} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1a) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u})}{2} \\
\text{z673} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\text{z674} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1b) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\text{z675} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1c) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u}) - \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u})}{2} \\
\text{z676} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1c) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{1u}) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_3^{(b)}(B_{2u})}{2} \\
\text{z677} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1d) &= -\frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2} \\
\text{z678} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1d) &= \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2} \\
\text{z679} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1e) &= \frac{-\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\text{z680} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1e) &= \frac{\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\text{z681} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1f) &= \frac{-\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z682}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1f) &= -\frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2} \\
\boxed{\text{z683}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 2) &= \frac{\sqrt{2} \left(\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) \right)}{8} \\
\boxed{\text{z684}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 2) &= \frac{\sqrt{2} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{1,2}^{(b)}(E_{1u}) \right)}{8} \\
\boxed{\text{z685}} \quad \mathbb{Q}_{9,1}^{(c)}(E_{2u}, 1) &= \frac{-\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z686}} \quad \mathbb{Q}_{9,2}^{(c)}(E_{2u}, 1) &= \frac{\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1)}{2} \\
\boxed{\text{z687}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, a) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a)}{2} \\
\boxed{\text{z688}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, a) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a)}{2} \\
\boxed{\text{z689}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, b) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b)}{2} \\
\boxed{\text{z690}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, b) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b)}{2} \\
\boxed{\text{z691}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, c) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) \right)}{8} \\
\boxed{\text{z692}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, c) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) \right)}{8} \\
\boxed{\text{z693}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, d) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) \right)}{8} \\
\boxed{\text{z694}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, d) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) \right)}{8}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z695}} \quad \mathbb{G}_{4,1}^{(c)}(E_{2u}, 1a) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z696}} \quad \mathbb{G}_{4,2}^{(c)}(E_{2u}, 1a) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{2u})}{2} \\
\boxed{\text{z697}} \quad \mathbb{G}_{4,1}^{(c)}(E_{2u}, 1b) &= \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a)}{2} \\
\boxed{\text{z698}} \quad \mathbb{G}_{4,2}^{(c)}(E_{2u}, 1b) &= \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, a) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, a)}{2} \\
\boxed{\text{z699}} \quad \mathbb{G}_{4,1}^{(c)}(E_{2u}, 1c) &= \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b)}{2} \\
\boxed{\text{z700}} \quad \mathbb{G}_{4,2}^{(c)}(E_{2u}, 1c) &= \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}, b) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}, b)}{2}
\end{aligned}$$

• **Sn;Sn_001_1** : 'Sn'-'Sn' bond-cluster

* bra: $\langle d_v |$, $\langle d_{xy} |$, $\langle d_{xz} |$, $\langle d_{yz} |$, $\langle d_u |$

* ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$

* wyckoff: **6a06g**

$$\begin{aligned}
\boxed{\text{z32}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) &= \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z33}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z34}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, c) &= \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_4^{(b)}(B_{2g}) \\
\boxed{\text{z35}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, a) &= \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z36}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z37}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, c) &= \frac{\sqrt{3} \left(2\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 2\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{6} \\
\boxed{\text{z38}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, a) &= \mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})
\end{aligned}$$

$$\begin{aligned}
\text{z39} \quad \mathbb{Q}_4^{(c)}(A_{1g}, b) &= \frac{\sqrt{3} \left(\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 2 \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 2 \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{6} \\
\text{z40} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 2) &= \frac{\sqrt{2} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2} \\
\text{z63} \quad \mathbb{G}_0^{(c)}(A_{1u}, a) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) + \mathbb{M}_1^{(a)}(A_{2g}) \mathbb{T}_1^{(b)}(A_{2u}) \right)}{3} \\
\text{z64} \quad \mathbb{G}_0^{(c)}(A_{1u}, b) &= \mathbb{M}_3^{(a)}(B_{1g}) \mathbb{T}_3^{(b)}(B_{1u}) \\
\text{z65} \quad \mathbb{G}_2^{(c)}(A_{1u}, a) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) + 2 \mathbb{M}_1^{(a)}(A_{2g}) \mathbb{T}_1^{(b)}(A_{2u}) \right)}{6} \\
\text{z66} \quad \mathbb{G}_2^{(c)}(A_{1u}, b) &= \frac{\sqrt{7} \left(\sqrt{2} \mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{2} \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{3} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_1^{(b)}(A_{2u}) \right)}{7} \\
\text{z67} \quad \mathbb{G}_2^{(c)}(A_{1u}, c) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{M}_{2,2}^{(b)}(E_{2u}) \right)}{2} \\
\text{z68} \quad \mathbb{G}_4^{(c)}(A_{1u}) &= \frac{\sqrt{7} \left(-\sqrt{6} \mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{6} \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) + 4 \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_1^{(b)}(A_{2u}) \right)}{14} \\
\text{z89} \quad \mathbb{Q}_6^{(c)}(A_{2g}, a) &= \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_4^{(b)}(B_{2g}) \\
\text{z90} \quad \mathbb{Q}_6^{(c)}(A_{2g}, b) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2} \\
\text{z91} \quad \mathbb{G}_1^{(c)}(A_{2g}) &= \frac{\sqrt{10} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 2 \mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 2 \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{10} \\
\text{z92} \quad \mathbb{G}_3^{(c)}(A_{2g}, a) &= \frac{\sqrt{10} \left(2 \mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2 \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{10} \\
\text{z93} \quad \mathbb{G}_3^{(c)}(A_{2g}, b) &= \frac{\sqrt{3} \left(\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 2 \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 2 \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z94}} \quad \mathbb{G}_5^{(c)}(A_{2g}) = \frac{\sqrt{3} \left(2\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{6}$$

$$\boxed{\text{z117}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, a) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z118}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, b) = \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z119}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, c) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) \right)}{2}$$

$$\boxed{\text{z120}} \quad \mathbb{Q}_3^{(c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{2}$$

$$\boxed{\text{z143}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2}$$

$$\boxed{\text{z144}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, b) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z145}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, c) = \frac{\sqrt{71} \left(-6\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 5\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 6\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 2\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 5\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) \right)}{142}$$

$$\boxed{\text{z146}} \quad \mathbb{Q}_6^{(c)}(B_{1g}) = \frac{\sqrt{71} \left(14\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 9\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 12\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 14\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 9\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 12\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) \right)}{710}$$

$$\boxed{\text{z147}} \quad \mathbb{G}_3^{(c)}(B_{1g}, a) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2}$$

$$\boxed{\text{z148}} \quad \mathbb{G}_3^{(c)}(B_{1g}, b) = \frac{-3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 2\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 2\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{10}$$

$$\boxed{\text{z180}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, a) = \frac{\sqrt{2} \left(-\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) \right)}{2}$$

$$\boxed{\text{z181}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, b) = \frac{-\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - 2\sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_1^{(b)}(A_{2u})}{4}$$

$$\boxed{\text{z182}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, c) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) - \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) \right)}{2}$$

$$\boxed{\text{z183}} \quad \mathbb{G}_4^{(c)}(B_{1u}) = \frac{-\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + 2\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_1^{(b)}(A_{2u})}{4}$$

$$\boxed{\text{z206}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_4^{(b)}(B_{2g})$$

$$\boxed{\text{z207}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, b) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_4^{(b)}(B_{2g})$$

$$\boxed{\text{z208}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2}$$

$$\boxed{\text{z209}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, d) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z210}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, e) = -\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_4^{(b)}(B_{2g})$$

$$\boxed{\text{z211}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, f) = \frac{\sqrt{71} \left(-6\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 2\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 5\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 6\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 5\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) \right)}{142}$$

$$\boxed{\text{z212}} \quad \mathbb{Q}_6^{(c)}(B_{2g}) = \frac{\sqrt{71} \left(14\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 9\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 12\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 14\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 9\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 12\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) \right)}{710}$$

$$\boxed{\text{z213}} \quad \mathbb{G}_3^{(c)}(B_{2g}, a) = \frac{-\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2}$$

$$\boxed{\text{z214}} \quad \mathbb{G}_3^{(c)}(B_{2g}, b) = \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 2\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 2\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{10}$$

$$\boxed{\text{z235}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, a) = -\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z236}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, b) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) \right)}{2}$$

$$\boxed{\text{z237}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, c) = \frac{-\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_1^{(b)}(A_{2u})}{4}$$

$$\boxed{\text{z238}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, d) = \mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z239}} \quad \mathbb{Q}_3^{(c)}(B_{2u}, e) = -\frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})\right)}{2}$$

$$\boxed{\text{z240}} \quad \mathbb{G}_4^{(c)}(B_{2u}) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + 2\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_1^{(b)}(A_{2u})}{4}$$

$$\boxed{\text{z329}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{2}$$

$$\boxed{\text{z330}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{2}$$

$$\boxed{\text{z331}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z332}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z333}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, c) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_4^{(b)}(B_{2g})}{2}$$

$$\boxed{\text{z334}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, c) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_4^{(b)}(B_{2g})}{2}$$

$$\boxed{\text{z335}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, d) = \frac{\sqrt{26}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})\right)}{26}$$

$$\boxed{\text{z336}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, d) = \frac{\sqrt{26}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})\right)}{26}$$

$$\boxed{\text{z337}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, e) = \frac{\sqrt{106}\left(-2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_4^{(b)}(B_{2g}) + 5\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_4^{(b)}(B_{2g})\right)}{106}$$

$$\boxed{\text{z338}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, e) = \frac{\sqrt{106}\left(-2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_4^{(b)}(B_{2g}) + 5\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_4^{(b)}(B_{2g})\right)}{106}$$

$$\boxed{\text{z339}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{1g}, f) = \frac{\sqrt{3} \left(\sqrt{10} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 4\sqrt{5} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \sqrt{10} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 4\sqrt{5} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 8 \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{70} \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{70} \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{48}$$

$$\boxed{\text{z340}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{1g}, f) = \frac{\sqrt{3} \left(-\sqrt{10} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 4\sqrt{5} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{10} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 4\sqrt{5} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 8 \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{70} \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{70} \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{48}$$

$$\boxed{\text{z341}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, a) = \frac{\sqrt{26} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 4\sqrt{3} \mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) \right)}{52}$$

$$\boxed{\text{z342}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, a) = \frac{\sqrt{26} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 4\sqrt{3} \mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) \right)}{52}$$

$$\boxed{\text{z343}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, b) = \frac{\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z344}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, b) = \frac{\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z345}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, c) = \frac{\sqrt{106} \left(5 \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_4^{(b)}(B_{2g}) + 2\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_4^{(b)}(B_{2g}) \right)}{106}$$

$$\boxed{\text{z346}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, c) = \frac{\sqrt{106} \left(5 \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_4^{(b)}(B_{2g}) + 2\sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_4^{(b)}(B_{2g}) \right)}{106}$$

$$\boxed{\text{z347}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{1g}, d) = \frac{\sqrt{1137} \left(-125\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 172 \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 125\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 172 \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 8\sqrt{5} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 29\sqrt{14} \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 29\sqrt{14} \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{18192}$$

$$\boxed{\text{z348}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{1g}, d) = \frac{\sqrt{1137} \left(125\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 172 \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 125\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 172 \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 8\sqrt{5} \mathbb{Q}_4^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 29\sqrt{14} \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 29\sqrt{14} \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{18192}$$

$$\boxed{\text{z349}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 1) = \frac{\sqrt{3} \left(\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{2} \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{2} \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{6}$$

$$\begin{aligned}
\boxed{\text{z350}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 1) &= \frac{\sqrt{3} \left(-\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{2}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{2}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{6} \\
\boxed{\text{z351}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{1g}, 2) &= \frac{\sqrt{379} \left(3\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 4\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 3\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 4\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 9\sqrt{70}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 17\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 17\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{2274} \\
\boxed{\text{z352}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{1g}, 2) &= \frac{\sqrt{379} \left(-3\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 4\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 3\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 4\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 9\sqrt{70}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 17\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 17\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{2274} \\
\boxed{\text{z353}} \quad \mathbb{G}_{1,1}^{(c)}(E_{1g}) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) \right)}{4} \\
\boxed{\text{z354}} \quad \mathbb{G}_{1,2}^{(c)}(E_{1g}) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) \right)}{4} \\
\boxed{\text{z355}} \quad \mathbb{G}_{3,1}^{(c)}(E_{1g}) &= \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 2\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 2\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{14}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{14}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{12} \\
\boxed{\text{z356}} \quad \mathbb{G}_{3,2}^{(c)}(E_{1g}) &= \frac{-3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 2\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 2\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{14}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{14}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{12} \\
\boxed{\text{z357}} \quad \mathbb{G}_{5,1}^{(c)}(E_{1g}, 1) &= \frac{\sqrt{3} \left(\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{6} \\
\boxed{\text{z358}} \quad \mathbb{G}_{5,2}^{(c)}(E_{1g}, 1) &= -\frac{\sqrt{3} \left(\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{6} \\
\boxed{\text{z461}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, a) &= \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_1^{(b)}(A_{2u}) - \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} \\
\boxed{\text{z462}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, a) &= \frac{-\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_1^{(b)}(A_{2u}) + \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} \\
\boxed{\text{z463}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, b) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2}
\end{aligned}$$

$$\begin{aligned}
\text{z464} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, b) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} \\
\text{z465} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, c) &= -\frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\text{z466} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, c) &= \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\text{z467} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}, d) &= \frac{-\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) - \sqrt{15}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \sqrt{15}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} \\
\text{z468} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}, d) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \sqrt{15}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \sqrt{15}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} \\
\text{z469} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, a) &= \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_1^{(b)}(A_{2u}) + \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - 6\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{12} \\
\text{z470} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, a) &= \frac{-\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_1^{(b)}(A_{2u}) + \sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + 6\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{12} \\
\text{z471} \quad \mathbb{Q}_{3,1}^{(c)}(E_{1u}, b) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) - \mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{8} \\
\text{z472} \quad \mathbb{Q}_{3,2}^{(c)}(E_{1u}, b) &= \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{8} \\
\text{z473} \quad \mathbb{Q}_{5,1}^{(c)}(E_{1u}, 1) &= \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} \\
\text{z474} \quad \mathbb{Q}_{5,2}^{(c)}(E_{1u}, 1) &= \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} \\
\text{z475} \quad \mathbb{G}_{2,1}^{(c)}(E_{1u}, a) &= \frac{\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_1^{(b)}(A_{2u}) + \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} \\
\text{z476} \quad \mathbb{G}_{2,2}^{(c)}(E_{1u}, a) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_1^{(b)}(A_{2u}) + \mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z477}} \quad \mathbb{G}_{2,1}^{(c)}(E_{1u}, b) &= \frac{\sqrt{7} \left(\sqrt{30} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) + 4\sqrt{3} \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_1^{(b)}(A_{2u}) - \sqrt{30} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) - 3\sqrt{2} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{42} \\
\boxed{\text{z478}} \quad \mathbb{G}_{2,2}^{(c)}(E_{1u}, b) &= \frac{\sqrt{7} \left(-4\sqrt{3} \mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_1^{(b)}(A_{2u}) - \sqrt{30} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) - \sqrt{30} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) + 3\sqrt{2} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{42} \\
\boxed{\text{z479}} \quad \mathbb{G}_{4,1}^{(c)}(E_{1u}) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{30} \mathbb{M}_{3,2}^{(a)}(E_{1g}) \mathbb{T}_1^{(b)}(A_{2u}) + \sqrt{3} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{5} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{28} \\
\boxed{\text{z480}} \quad \mathbb{G}_{4,2}^{(c)}(E_{1u}) &= \frac{\sqrt{7} \left(-\sqrt{30} \mathbb{M}_{3,1}^{(a)}(E_{1g}) \mathbb{T}_1^{(b)}(A_{2u}) + \sqrt{3} \mathbb{M}_{3,1}^{(a)}(E_{2g}) \mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{3} \mathbb{M}_{3,2}^{(a)}(E_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) - 2\sqrt{5} \mathbb{M}_3^{(a)}(A_{2g}) \mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{28} \\
\boxed{\text{z583}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z584}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z585}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, b) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_{2g}) \mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z586}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, b) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_{2g}) \mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z587}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, c) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_4^{(b)}(B_{2g})}{2} \\
\boxed{\text{z588}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, c) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_4^{(b)}(B_{2g})}{2} \\
\boxed{\text{z589}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, d) &= \frac{\sqrt{5} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 2\mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{2g}) \right)}{10} \\
\boxed{\text{z590}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, d) &= \frac{\sqrt{5} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 2\mathbb{Q}_2^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_{2g}) \right)}{10} \\
\boxed{\text{z591}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, e) &= -\frac{\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_{1g}) \mathbb{Q}_4^{(b)}(B_{2g})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z592}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, e) &= -\frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_4^{(b)}(B_{2g})}{2} \\
\boxed{\text{z593}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}, f) &= \frac{\sqrt{111}\left(\sqrt{5}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{70}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{5}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{70}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{2}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{35}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \sqrt{35}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})\right)}{222} \\
\boxed{\text{z594}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}, f) &= \frac{\sqrt{111}\left(-\sqrt{5}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{70}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{5}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \sqrt{70}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{2}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{35}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{35}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})\right)}{222} \\
\boxed{\text{z595}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1a) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z596}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1a) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2} \\
\boxed{\text{z597}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1b) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z598}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1b) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z599}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 1c) &= \frac{-\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} \\
\boxed{\text{z600}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 1c) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} \\
\boxed{\text{z601}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2a) &= \frac{\sqrt{10}\left(-\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})\right)}{10} \\
\boxed{\text{z602}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2a) &= \frac{\sqrt{10}\left(\mathbb{Q}_{2,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + \mathbb{Q}_{2,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + \sqrt{3}\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})\right)}{10} \\
\boxed{\text{z603}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z604}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z605}} \quad \mathbb{Q}_{4,1}^{(c)}(E_{2g}, 2c) &= \frac{\sqrt{30747} \left(-137\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 26\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 137\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 26\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 94\sqrt{10}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 85\sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 85\sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) \right)}{122988} \\
\boxed{\text{z606}} \quad \mathbb{Q}_{4,2}^{(c)}(E_{2g}, 2c) &= \frac{\sqrt{30747} \left(137\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) + 26\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 137\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 26\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 94\sqrt{10}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 85\sqrt{7}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 85\sqrt{7}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) \right)}{122988} \\
\boxed{\text{z607}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{2g}, 1) &= \frac{\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{2}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - \sqrt{2}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g})}{6} \\
\boxed{\text{z608}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{2g}, 1) &= \frac{-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{2}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - \sqrt{2}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g})}{6} \\
\boxed{\text{z609}} \quad \mathbb{Q}_{6,1}^{(c)}(E_{2g}, 2) &= \frac{\sqrt{831} \left(-34\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 33\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 34\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 33\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 14\sqrt{35}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 6\sqrt{2}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 6\sqrt{2}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) \right)}{8310} \\
\boxed{\text{z610}} \quad \mathbb{Q}_{6,2}^{(c)}(E_{2g}, 2) &= \frac{\sqrt{831} \left(34\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 33\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + 34\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 33\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 14\sqrt{35}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 6\sqrt{2}\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 6\sqrt{2}\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) \right)}{8310} \\
\boxed{\text{z611}} \quad \mathbb{G}_{3,1}^{(c)}(E_{2g}) &= \frac{\sqrt{3} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) + 6\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 6\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 2\sqrt{70}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g}) + 9\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 9\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) \right)}{60} \\
\boxed{\text{z612}} \quad \mathbb{G}_{3,2}^{(c)}(E_{2g}) &= \frac{\sqrt{3} \left(\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) - 6\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,2}^{(b)}(E_{2g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 6\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1)\mathbb{Q}_{2,1}^{(b)}(E_{2g}) - 2\sqrt{70}\mathbb{Q}_4^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g}) - 9\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{1g}) - 9\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_{2,2}^{(b)}(E_{1g}) \right)}{60} \\
\boxed{\text{z701}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, a) &= -\frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z702}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, a) &= \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z703}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, b) &= -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} \\
\boxed{\text{z704}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, b) &= \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} \\
\boxed{\text{z705}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, c) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - 2\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_1^{(b)}(A_{2u}) + 3\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + 3\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{12} \\
\boxed{\text{z706}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, c) &= \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_1^{(b)}(A_{2u}) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - 3\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + 3\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{12} \\
\boxed{\text{z707}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, d) &= \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z708}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, d) &= -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_3^{(b)}(B_{1u})}{2} \\
\boxed{\text{z709}} \quad \mathbb{Q}_{3,1}^{(c)}(E_{2u}, e) &= \frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} \\
\boxed{\text{z710}} \quad \mathbb{Q}_{3,2}^{(c)}(E_{2u}, e) &= -\frac{\sqrt{2}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} \\
\boxed{\text{z711}} \quad \mathbb{Q}_{5,1}^{(c)}(E_{2u}, 1) &= \frac{-\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u})}{2} \\
\boxed{\text{z712}} \quad \mathbb{Q}_{5,2}^{(c)}(E_{2u}, 1) &= \frac{\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{M}_{2,2}^{(b)}(E_{2u}) + \mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{M}_{2,1}^{(b)}(E_{2u})}{2} \\
\boxed{\text{z713}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, a) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} \\
\boxed{\text{z714}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, a) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \mathbb{M}_{1,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z715}} \quad \mathbb{G}_{2,1}^{(c)}(E_{2u}, b) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \sqrt{30}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_1^{(b)}(A_{2u}) + 3\sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + 3\sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{42} \\
\boxed{\text{z716}} \quad \mathbb{G}_{2,2}^{(c)}(E_{2u}, b) &= \frac{\sqrt{7} \left(\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \sqrt{30}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_1^{(b)}(A_{2u}) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - 3\sqrt{5}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + 3\sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{42} \\
\boxed{\text{z717}} \quad \mathbb{G}_{4,1}^{(c)}(E_{2u}, 1) &= \frac{-\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u})}{2} \\
\boxed{\text{z718}} \quad \mathbb{G}_{4,2}^{(c)}(E_{2u}, 1) &= \frac{\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u})}{2} \\
\boxed{\text{z719}} \quad \mathbb{G}_{4,1}^{(c)}(E_{2u}, 2) &= \frac{\sqrt{7} \left(\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) + 2\sqrt{6}\mathbb{M}_{3,2}^{(a)}(E_{2g})\mathbb{T}_1^{(b)}(A_{2u}) - \mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) \right)}{28} \\
\boxed{\text{z720}} \quad \mathbb{G}_{4,2}^{(c)}(E_{2u}, 2) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_{1g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) - 2\sqrt{6}\mathbb{M}_{3,1}^{(a)}(E_{2g})\mathbb{T}_1^{(b)}(A_{2u}) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) + \mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_{1u}) - \mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_{1u}) \right)}{28}
\end{aligned}$$

Atomic SAMB

- bra: $\langle d_v |, \langle d_{xy} |, \langle d_{xz} |, \langle d_{yz} |, \langle d_u |$
- ket: $|d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$$

$$\boxed{\text{x2}} \quad \mathbb{Q}_2^{(a)}(A_{1g}) = \begin{bmatrix} -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{Q}_4^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{35} \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{Q}_4^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x5}} \quad \mathbb{Q}_4^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x6}} \quad \mathbb{Q}_{2,1}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{Q}_{2,2}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x8}} \quad \mathbb{Q}_{4,1}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{\sqrt{21}}{7} \\ 0 & 0 & 0 & \frac{\sqrt{21}}{7} & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{Q}_{4,2}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}}{7} \\ 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{7} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x10}} \quad \mathbb{Q}_{2,1}^{(a)}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x11}} \quad \mathbb{Q}_{2,2}^{(a)}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x12}} \quad \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x13}} \quad \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x14}} \quad \mathbb{Q}_{4,1}^{(a)}(E_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{7} & 0 \\ \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x15}} \quad \mathbb{Q}_{4,2}^{(a)}(E_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{7} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x16}} \quad \mathbb{M}_1^{(a)}(A_{2g}) = \begin{bmatrix} 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x17}} \quad \mathbb{M}_3^{(a)}(A_{2g}) = \begin{bmatrix} 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x18}} \quad \mathbb{M}_3^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x19}} \quad \mathbb{M}_3^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x20}} \quad \mathbb{M}_{1,1}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x21}} \quad \mathbb{M}_{1,2}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & \frac{\sqrt{30}i}{10} \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{10} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x22}} \quad \mathbb{M}_{3,1}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 \end{bmatrix}$$

$$\boxed{\text{x23}} \quad \mathbb{M}_{3,2}^{(a)}(E_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x24}} \quad \mathbb{M}_{3,1}^{(a)}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x25}} \quad \mathbb{M}_{3,2}^{(a)}(E_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$$

Cluster SAMB

- Site cluster

** Wyckoff: 2c

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_{1g}) = \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right]$$

$$\boxed{\text{y2}} \quad \mathbb{Q}_3^{(s)}(B_{1u}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

** Wyckoff: 6h

$$\boxed{\text{y3}} \quad \mathbb{Q}_0^{(s)}(A_{1g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y4}} \quad \mathbb{Q}_3^{(s)}(B_{1u}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y5}} \quad \mathbb{Q}_{1,1}^{(s)}(E_{1u}) = \left[0, -\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2} \right]$$

$$\boxed{\text{y6}} \quad \mathbb{Q}_{1,2}^{(s)}(E_{1u}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y7}} \quad \mathbb{Q}_{2,1}^{(s)}(E_{2g}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y8}} \quad \mathbb{Q}_{2,2}^{(s)}(E_{2g}) = \left[0, \frac{1}{2}, -\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2} \right]$$

• Bond cluster

** Wyckoff: **6b@6h**

$$\boxed{\text{y9}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y10}} \quad \mathbb{M}_1^{(b)}(A_{2g}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y11}} \quad \mathbb{Q}_3^{(b)}(B_{1u}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y12}} \quad \mathbb{T}_3^{(b)}(B_{2u}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y13}} \quad \mathbb{Q}_{1,1}^{(b)}(E_{1u}) = \left[0, -\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2} \right]$$

$$\boxed{\text{y14}} \quad \mathbb{Q}_{1,2}^{(b)}(E_{1u}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y15}} \quad \mathbb{T}_{1,1}^{(b)}(E_{1u}) = \left[\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y16}} \quad \mathbb{T}_{1,2}^{(b)}(E_{1u}) = \left[0, \frac{i}{2}, -\frac{i}{2}, 0, -\frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y17}} \quad \mathbb{Q}_{2,1}^{(b)}(E_{2g}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y18}} \quad \mathbb{Q}_{2,2}^{(b)}(E_{2g}) = \left[0, \frac{1}{2}, -\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2} \right]$$

$$\boxed{\text{y19}} \quad \mathbb{T}_{2,1}^{(b)}(E_{2g}) = \left[0, \frac{i}{2}, -\frac{i}{2}, 0, \frac{i}{2}, -\frac{i}{2} \right]$$

$$\boxed{\text{y20}} \quad \mathbb{T}_{2,2}^{(b)}(E_{2g}) = \left[-\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

** Wyckoff: **6a@6g**

$$\boxed{\text{y21}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y22}} \quad \mathbb{T}_1^{(b)}(A_{2u}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y23}} \quad \mathbb{T}_3^{(b)}(B_{1u}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y24}} \quad \mathbb{Q}_4^{(b)}(B_{2g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y25}} \quad \mathbb{Q}_{2,1}^{(b)}(E_{1g}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y26}} \quad \mathbb{Q}_{2,2}^{(b)}(E_{1g}) = \left[0, \frac{1}{2}, -\frac{1}{2}, 0, -\frac{1}{2}, \frac{1}{2} \right]$$

$$\boxed{\text{y27}} \quad \mathbb{T}_{1,1}^{(b)}(E_{1u}) = \left[0, -\frac{i}{2}, \frac{i}{2}, 0, \frac{i}{2}, -\frac{i}{2} \right]$$

$$\boxed{\text{y28}} \quad \mathbb{T}_{1,2}^{(b)}(E_{1u}) = \left[\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y29}} \quad \mathbb{Q}_{2,1}^{(b)}(E_{2g}) = \left[\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y30}} \quad \mathbb{Q}_{2,2}^{(b)}(E_{2g}) = \left[0, \frac{1}{2}, -\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2} \right]$$

$$\boxed{\text{y31}} \quad \mathbb{M}_{2,1}^{(b)}(E_{2u}) = \left[0, -\frac{i}{2}, \frac{i}{2}, 0, -\frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y32}} \quad \mathbb{M}_{2,2}^{(b)}(E_{2u}) = \left[\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6} \right]$$

** Wyckoff: 12a012j

$$\boxed{\text{y33}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y34}} \quad \mathbb{T}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y35}} \quad \mathbb{M}_1^{(b)}(A_{2g}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y36}} \quad \mathbb{Q}_6^{(b)}(A_{2g}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y37}} \quad \mathbb{Q}_3^{(b)}(B_{1u}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y38}} \quad \mathbb{T}_3^{(b)}(B_{1u}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y39}} \quad \mathbb{Q}_3^{(b)}(B_{2u}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y40}} \quad \mathbb{T}_3^{(b)}(B_{2u}) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y41}} \quad \mathbb{Q}_{1,1}^{(b)}(E_{1u}) = \left[\frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, -\frac{\sqrt{42}}{84}, -\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, -\frac{\sqrt{42}}{84}, \frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, -\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21} \right]$$

$$\boxed{\text{y42}} \quad \mathbb{Q}_{1,2}^{(b)}(E_{1u}) = \left[\frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14}, -\frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{28}, \frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14}, -\frac{3\sqrt{14}}{28}, \frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14} \right]$$

$$\boxed{\text{y43}} \quad \mathbb{T}_{1,1}^{(b)}(E_{1u}, a) = \left[\frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, -\frac{\sqrt{42}i}{84}, -\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{84}, \frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, -\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21} \right]$$

$$\boxed{\text{y44}} \quad \mathbb{T}_{1,2}^{(b)}(E_{1u}, a) = \left[\frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14}, -\frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14}, \frac{3\sqrt{14}i}{28}, \frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14}, -\frac{3\sqrt{14}i}{28}, \frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14} \right]$$

$$\boxed{\text{y45}} \quad \mathbb{T}_{1,1}^{(b)}(E_{1u}, b) = \left[\frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14}, -\frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14}, \frac{3\sqrt{14}i}{28}, -\frac{3\sqrt{14}i}{28}, \frac{\sqrt{14}i}{28}, \frac{\sqrt{14}i}{14}, \frac{3\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{28}, -\frac{\sqrt{14}i}{14} \right]$$

$$\boxed{\text{y46}} \quad \mathbb{T}_{1,2}^{(b)}(E_{1u}, b) = \left[-\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, \frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, -\frac{\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{84}, \frac{5\sqrt{42}i}{84}, -\frac{\sqrt{42}i}{21}, \frac{\sqrt{42}i}{84}, -\frac{5\sqrt{42}i}{84}, \frac{\sqrt{42}i}{21} \right]$$

$$\boxed{\text{y47}} \quad \mathbb{Q}_{5,1}^{(b)}(E_{1u}, 1) = \left[\frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14}, -\frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{28}, -\frac{3\sqrt{14}}{28}, \frac{\sqrt{14}}{28}, \frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{28}, -\frac{\sqrt{14}}{28}, -\frac{\sqrt{14}}{14} \right]$$

$$\boxed{\text{y48}} \quad \mathbb{Q}_{5,2}^{(b)}(E_{1u}, 1) = \left[-\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, \frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, -\frac{\sqrt{42}}{84}, -\frac{\sqrt{42}}{84}, \frac{5\sqrt{42}}{84}, -\frac{\sqrt{42}}{21}, \frac{\sqrt{42}}{84}, -\frac{5\sqrt{42}}{84}, \frac{\sqrt{42}}{21} \right]$$

$$\boxed{\text{y49}} \quad \mathbb{Q}_{2,1}^{(b)}(E_{2g}) = \left[\frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42} \right]$$

$$\boxed{\text{y50}} \quad \mathbb{Q}_{2,2}^{(b)}(E_{2g}) = \left[-\frac{5\sqrt{2}}{28}, \frac{2\sqrt{2}}{7}, -\frac{3\sqrt{2}}{28}, -\frac{5\sqrt{2}}{28}, \frac{2\sqrt{2}}{7}, -\frac{3\sqrt{2}}{28}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7} \right]$$

$$\boxed{\text{y51}} \quad \mathbb{T}_{2,1}^{(b)}(E_{2g}, a) = \left[\frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42} \right]$$

$$\boxed{\text{y52}} \quad \mathbb{T}_{2,2}^{(b)}(E_{2g}, a) = \left[-\frac{5\sqrt{2}i}{28}, \frac{2\sqrt{2}i}{7}, -\frac{3\sqrt{2}i}{28}, -\frac{5\sqrt{2}i}{28}, \frac{2\sqrt{2}i}{7}, -\frac{3\sqrt{2}i}{28}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7} \right]$$

$$\boxed{\text{y53}} \quad \mathbb{T}_{2,1}^{(b)}(E_{2g}, b) = \left[\frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7}, \frac{3\sqrt{2}i}{28}, \frac{5\sqrt{2}i}{28}, -\frac{2\sqrt{2}i}{7} \right]$$

$$\boxed{\text{y54}} \quad \mathbb{T}_{2,2}^{(b)}(E_{2g}, b) = \left[\frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{11\sqrt{6}i}{84}, \frac{\sqrt{6}i}{42}, -\frac{13\sqrt{6}i}{84}, \frac{13\sqrt{6}i}{84}, -\frac{11\sqrt{6}i}{84}, -\frac{\sqrt{6}i}{42}, \frac{13\sqrt{6}i}{84}, -\frac{11\sqrt{6}i}{84}, -\frac{\sqrt{6}i}{42} \right]$$

$$\boxed{\text{y55}} \quad \mathbb{Q}_{4,1}^{(b)}(E_{2g}, 1) = \left[\frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7}, \frac{3\sqrt{2}}{28}, \frac{5\sqrt{2}}{28}, -\frac{2\sqrt{2}}{7} \right]$$

$$\boxed{\text{y56}} \quad \mathbb{Q}_{4,2}^{(b)}(E_{2g}, 1) = \left[\frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{11\sqrt{6}}{84}, \frac{\sqrt{6}}{42}, -\frac{13\sqrt{6}}{84}, \frac{13\sqrt{6}}{84}, -\frac{11\sqrt{6}}{84}, -\frac{\sqrt{6}}{42}, \frac{13\sqrt{6}}{84}, -\frac{11\sqrt{6}}{84}, -\frac{\sqrt{6}}{42} \right]$$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	Mn	$ d_v\rangle, d_{xy}\rangle, d_{xz}\rangle, d_{yz}\rangle, d_u\rangle$
2	Sn	$ d_v\rangle, d_{xy}\rangle, d_{xz}\rangle, d_{yz}\rangle, d_u\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	Mn	Mn	[1]	[d]	[d]
2	Mn	Sn	[1]	[d]	[d]
3	Sn	Sn	[1]	[d]	[d]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'Mn' (#1) site cluster (6h), mm2

SL	position (<i>s</i>)	mapping
1	[0.83880, 0.67760, 0.25000]	[1,11,16,20]
2	[0.32240, 0.16120, 0.25000]	[2,10,17,19]
3	[0.83880, 0.16120, 0.25000]	[3,12,18,21]
4	[0.16120, 0.32240, 0.75000]	[4,8,13,23]
5	[0.67760, 0.83880, 0.75000]	[5,7,14,22]
6	[0.16120, 0.83880, 0.75000]	[6,9,15,24]

Table 8: 'Sn' (#2) site cluster (2c), -6m2

SL	position (<i>s</i>)	mapping
1	[0.33333, 0.66667, 0.25000]	[1,2,3,10,11,12,16,17,18,19,20,21]
2	[0.66667, 0.33333, 0.75000]	[4,5,6,7,8,9,13,14,15,22,23,24]

— Bond in Unit Cell —

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 9: 1-th 'Mn'-'Mn' [1] (#1) bond cluster (6b@6h), ND, $|\mathbf{v}|=0.4836$ (cartesian)

SL	vector ($\mathbf{v}$)	center ($\mathbf{c}$)	mapping	head	tail	$\mathbf{R}$ (primitive)
1	[-0.48360, 0.00000, 0.00000]	[0.08060, 0.16120, 0.25000]	[1,-11,16,-20]	(3,1)	(2,1)	[1,0,0]
2	[0.00000,-0.48360, 0.00000]	[0.83880, 0.91940, 0.25000]	[2,-10,17,-19]	(1,1)	(3,1)	[0,1,0]
3	[0.48360, 0.48360, 0.00000]	[0.08060, 0.91940, 0.25000]	[3,-12,18,-21]	(2,1)	(1,1)	[-1,-1,0]
4	[0.48360, 0.00000, 0.00000]	[0.91940, 0.83880, 0.75000]	[4,-8,13,-23]	(6,1)	(5,1)	[-1,0,0]
5	[0.00000, 0.48360, 0.00000]	[0.16120, 0.08060, 0.75000]	[5,-7,14,-22]	(4,1)	(6,1)	[0,-1,0]
6	[-0.48360,-0.48360, 0.00000]	[0.91940, 0.08060, 0.75000]	[6,-9,15,-24]	(5,1)	(4,1)	[1,1,0]

Table 10: 1-th 'Mn'-'Sn' [1] (#2) bond cluster (12a@12j), D, $|\mathbf{v}|=0.50009$ (cartesian)

SL	vector ($\mathbf{v}$)	center ($\mathbf{c}$)	mapping	head	tail	$\mathbf{R}$ (primitive)
1	[-0.49453, 0.01093, 0.00000]	[0.08607, 0.67213, 0.25000]	[1,16]	(1,1)	(1,1)	[1,0,0]
2	[-0.01093,-0.50547, 0.00000]	[0.32787, 0.41393, 0.25000]	[2,17]	(2,1)	(1,1)	[0,0,0]
3	[0.50547, 0.49453, 0.00000]	[0.58607, 0.91393, 0.25000]	[3,18]	(3,1)	(1,1)	[0,-1,0]
4	[0.49453,-0.01093, 0.00000]	[0.91393, 0.32787, 0.75000]	[4,13]	(4,1)	(2,1)	[-1,0,0]
5	[0.01093, 0.50547, 0.00000]	[0.67213, 0.58607, 0.75000]	[5,14]	(5,1)	(2,1)	[0,0,0]
6	[-0.50547,-0.49453, 0.00000]	[0.41393, 0.08607, 0.75000]	[6,15]	(6,1)	(2,1)	[0,1,0]
7	[0.01093,-0.49453, 0.00000]	[0.67213, 0.08607, 0.75000]	[7,22]	(5,1)	(2,1)	[0,1,0]
8	[-0.50547,-0.01093, 0.00000]	[0.41393, 0.32787, 0.75000]	[8,23]	(4,1)	(2,1)	[0,0,0]
9	[0.49453, 0.50547, 0.00000]	[0.91393, 0.58607, 0.75000]	[9,24]	(6,1)	(2,1)	[-1,0,0]

continued ...

Table 10

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
10	[-0.01093, 0.49453, 0.00000]	[0.32787, 0.91393, 0.25000]	[10,19]	(2,1)	(1,1)	[0,-1,0]
11	[0.50547, 0.01093, 0.00000]	[0.58607, 0.67213, 0.25000]	[11,20]	(1,1)	(1,1)	[0,0,0]
12	[-0.49453,-0.50547, 0.00000]	[0.08607, 0.41393, 0.25000]	[12,21]	(3,1)	(1,1)	[1,0,0]

Table 11: 1-th 'Sn'-'Sn' [1] (#3) bond cluster (6a06g), ND, $|\boldsymbol{v}|=0.76376$ (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.33333, 0.66667,-0.50000]	[0.50000, 0.00000, 0.00000]	[1,-8,-13,20]	(2,1)	(1,1)	[0,-1,1]
2	[-0.66667,-0.33333,-0.50000]	[0.00000, 0.50000, 0.00000]	[2,-7,-14,19]	(2,1)	(1,1)	[1,0,1]
3	[0.33333,-0.33333,-0.50000]	[0.50000, 0.50000, 0.00000]	[3,-9,-15,21]	(2,1)	(1,1)	[0,0,1]
4	[-0.33333,-0.66667,-0.50000]	[0.50000, 0.00000, 0.50000]	[4,-11,-16,23]	(1,1)	(2,1)	[0,1,0]
5	[0.66667, 0.33333,-0.50000]	[0.00000, 0.50000, 0.50000]	[5,-10,-17,22]	(1,1)	(2,1)	[-1,0,0]
6	[-0.33333, 0.33333,-0.50000]	[0.50000, 0.50000, 0.50000]	[6,-12,-18,24]	(1,1)	(2,1)	[0,0,0]